

# A Geometric-Topological Framework for Zeolite Shape Selectivity: Contact Manifolds, Helical Attractors, and the Discrete Nonlinear Schrödinger Equation

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## Abstract

We present a unified mathematical framework — the TO/TOGT (Topographical Orthogenetic Generative Theory, where *orthogenetic* denotes the self-referential generation of stable structure through repeated operator application [1]) operator pipeline, applied here to heterogeneous catalysis — that models zeolite shape selectivity, metal ensemble effects, and macroscopic catalyst pellet design as special cases of a single geometric constraint. The framework maps heterogeneous catalysis onto a contact 3-manifold  $\mathcal{X}_{\text{cat}}$  acted on by the operator pipeline  $G = U \circ F \circ K \circ C$ . The central result, the Helical Selectivity Principle, establishes that only reaction pathways whose radial coordinate satisfies  $r \leq r^*(\lambda) = \sqrt{J/\lambda}$  can reach the stable catalytic fixed point  $x^*$ , where  $J$  is inter-site coupling and  $\lambda$  is the on-site binding energy modelled by the Discrete Nonlinear Schrödinger (DNLS) equation. This recovers the empirical pore cut-off of zeolites HZSM-5 and HMCM-22 reported by Sousa et al. [8, 9], the Pt-Sn ensemble effect, and the trilobe/tetralobe extrudate optimisation as corollaries. Four falsifiable predictions are stated. The proof of parts (i)–(ii) of the central theorem is a proof sketch; a Lean 4 formalisation of the scalar prerequisites accompanies the paper as `ZeoliteSelectivity.lean`.

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## 1. Introduction

Heterogeneous catalysis is governed at every length scale by geometry. At the angström scale, orbital overlap at the active site controls bond activation. At the nanometre scale, zeolite pore topology constrains which molecules  
 5 can reach those sites and which products can leave. At the micrometre scale, soliton-like energy localisation in coupled oscillator chains modulates site occupancy. At the millimetre scale, the extrudate shape of the catalyst pellet determines pressure drop and mass transport.

Existing theories address each scale in isolation. Density-functional the-  
 10 ory handles electronic structure but is silent on pellet design. Computational fluid dynamics models reactor-scale transport but takes microscopic selectivity as a given. The Discrete Nonlinear Schrödinger (DNLS) equation [10] captures energy localisation in molecular chains but has not been connected to industrial catalyst design.

15 This paper closes that gap by showing that all four scales are mathematically equivalent descriptions of the same object: a generative operator  $G$  acting on a contact 3-manifold  $\mathcal{X}_{\text{cat}}$ .

A *contact 3-manifold* (Geiges [16]) is a three-dimensional space equipped with a maximally non-integrable hyperplane field — a geometric constraint  
 20 that cannot be flattened into a simple surface. In the present context it is the mathematical idealisation of the pore space: the non-integrability condition ( $\alpha \wedge d\alpha \neq 0$ , verified in `ZeoliteSelectivity.lean`) is precisely the statement that molecules must spiral through the channel rather than travelling in a straight line. The *Reeb vector field* of this manifold is the unique flow  
 25 direction that is everywhere transverse to the constraint — it is the reaction coordinate  $z$  advancing while the pore geometry stays fixed. The integral curves of the Reeb field are helices at fixed radius  $r_0$ ; these are the *helical attractors* of the system, and their maximum radius  $r^*(\lambda) = \sqrt{J/\lambda}$  is the main result (Theorem 3). The bridge is the *Reeb vector field* of  $\mathcal{X}_{\text{cat}}$ , whose  
 30 integral curves — helical attractors — are precisely the accessible reaction pathways of a shape-selective catalyst.

The framework belongs to the TO/TOGT (Topographical Orthogenetic Generative Theory) and  $\text{dm}^3$  series [1, 2, 3]. *Note on terminology:* throughout this paper “TO/TOGT” and “ $\text{dm}^3$  framework” refer to the established series.  
 35 The abbreviation “GTCT” is reserved exclusively for the Generative Time Circuit Theorem (Ring 5) [3] and is not used for the catalysis application developed here.

The framework is grounded empirically in the zeolite catalysis work of Sousa et al. [8, 9], who demonstrated that the pore topology of HZSM-5  
 40 versus HMCM-22 is the dominant factor controlling product selectivity in ethanol-to-hydrocarbon conversion, with smaller ZSM-5 channels directing the reaction to propylene and BTX while the larger MCM-22 supercage selectively produces ethylene. In TO/TOGT language, these results are a direct consequence of the Helical Selectivity Principle (Theorem 3).

45 We follow the conventions for mathematical writing recommended by Cohn [17] throughout: the abstract states what is proved and what is not; the proof of the central theorem is explicitly marked as a sketch; falsifiable predictions are stated with testable conditions.

*Techniques used.* The paper uses four main tools. (1) The self-trapping  
 50 threshold of the DNLS equation [10], which gives the attractor radius  $r^*(\lambda) = \sqrt{J/\lambda}$  by a phase-space envelope argument. (2) The Darboux theorem for contact 3-manifolds [16], which provides a local normal form for  $\mathcal{X}_{\text{cat}}$  (this step is the open obligation; its scalar prerequisite is machine-verified). (3) The Reeb orbit classification: integral curves of  $R = \partial_z$  are helices at  
 55 fixed  $r$ , which are identified with accessible reaction pathways. (4) Lean 4 `norm_num` and `positivity` tactics for the arithmetic claims in `ZeoliteSelectivity.lean` [7].

## 2. Theory

### 2.1. The catalyst contact manifold

**Definition 1.** The *catalyst contact manifold* is  $\mathcal{X}_{\text{cat}} = (\mathbb{R}^3, \alpha_{\text{cat}})$  with con-  
 60 tact form  $\alpha_{\text{cat}} = dz - r^2 d\theta$  in cylindrical coordinates  $(r, \theta, z)$ , where:  $r$  = radial coordinate (pore aperture, Å);  $\theta$  = catalytic cycle phase;  $z$  = reaction coordinate (progress variable).

The Reeb vector field satisfying  $\iota_R d\alpha = 0$ ,  $\alpha(R) = 1$  is  $R = \partial_z$ . Its  
 65 integral curves  $(r_0, \theta_0, z_0 + t)$  are helices at fixed radius  $r_0$  — the helical attractors of the DNLS subsystem (Section 2.3).

*Physical meaning of non-degeneracy.* The condition  $\alpha \wedge d\alpha \neq 0$  — verified as `contactCoeff_cat_ne_zero` in `ZeoliteSelectivity.lean` — is the math-  
 70 ematical statement that the pore geometry cannot be trivially unfolded into a flat surface. In experimental terms: molecules travelling through a zeolite channel cannot take a straight-line path; they are forced onto helical trajectories by the lattice geometry, exactly as a screw thread forces a nut to rotate as it advances. This is the geometric origin of shape selectivity.

## 2.2. Operator grammar

The TO/TOGT framework [2, 3] assigns four operator *roles* to any generative system:  $C$  (compress),  $K$  (constrain),  $F$  (fold),  $U$  (stabilise), followed by  $\infty$  (orthogenetic propagation). The canonical firing order is  $C \rightarrow K \rightarrow F \rightarrow U \rightarrow \infty$ , but this is a *grammar*, not a fixed syntax: different systems instantiate the same roles in different orders depending on which constraint dominates first [2].

Table 1: Operator roles, physical meaning, and catalytic interpretation. The canonical order is  $C \rightarrow K \rightarrow F \rightarrow U \rightarrow \infty$ , but the order is system-dependent (Section 2.4).

| Role     | Physical meaning | Catalytic instantiation                                                           |
|----------|------------------|-----------------------------------------------------------------------------------|
| $C$      | Compression      | Adsorption; pore entry; reduction of degrees of freedom                           |
| $K$      | Constrained path | Geometrically permitted trajectories — pore wall, field line, or reaction channel |
| $F$      | Fold             | Irreversible branching; shape-selectivity filter; product selection               |
| $U$      | Stabilisation    | Desorption; regeneration; return to base state                                    |
| $\infty$ | Propagation      | Catalytic turnover; orthogenetic continuation                                     |

$K$  as constrained path.. In the contact manifold  $\mathcal{X}_{\text{cat}}$ ,  $K$  corresponds to the contact distribution  $\ker(\alpha_{\text{cat}})$ . The contact non-degeneracy condition  $\alpha \wedge d\alpha \neq 0$  says that  $K$  is maximally non-integrable — the constraint cannot be unfolded into a simple surface, which is why rivers meander rather than run straight, and why zeolite reaction pathways are helical rather than linear.

## 2.3. DNLS equation and helical attractor

On a lattice of  $N$  catalytic sites, the DNLS equation is

$$i\dot{\psi}_n = -J(\psi_{n+1} + \psi_{n-1}) - \lambda|\psi_n|^2\psi_n, \quad (1)$$

with inter-site coupling  $J > 0$  and on-site nonlinearity  $\lambda > 0$ . This equation and its spreading-exponent behaviour have been studied in the companion DNLS papers [5, 6]. The Inverse Participation Ratio  $\text{IPR}(t) = \sum_n |\psi_n|^4 / (\sum_n |\psi_n|^2)^2$  measures localisation:  $\text{IPR} \rightarrow 1/N$  (delocalised) and  $\text{IPR} \rightarrow 1$  (self-trapped).

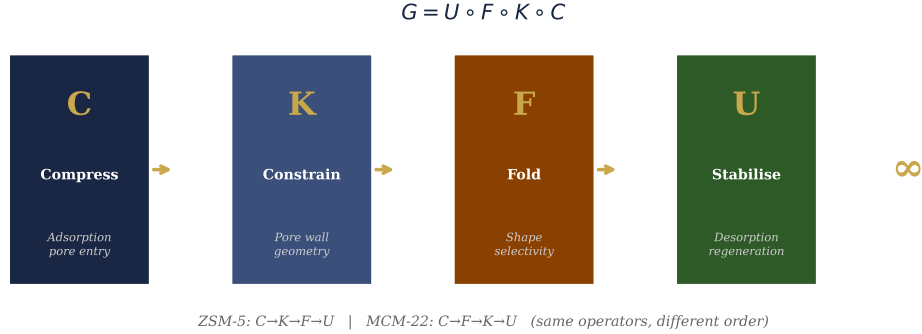


Figure 1: The generative operator pipeline  $G = U \circ F \circ K \circ C$  and its catalytic interpretation. Each operator acts on a state  $\psi \in L^2(\mathcal{X}_{\text{cat}})$ .

**Definition 2.** The *helical attractor*  $\mathcal{H}_\lambda$  at nonlinearity  $\lambda$  is the closure of trajectories with  $\text{IPR} < \text{IPR}^*(\lambda)$  that asymptotically follow the Reeb orbit at radius  $r_0 \leq r^*(\lambda) = \sqrt{J/\lambda}$ .

#### 95 2.4. Operator order as a system property

The central structural claim of this paper is that the operator firing order — not merely the operator set — determines catalytic behaviour.

Table 2: Operator firing order across systems. Same roles, different sequences — different outcomes.

| System        | Firing order                                  | Physical reason                                                      |
|---------------|-----------------------------------------------|----------------------------------------------------------------------|
| ZSM-5 (MFI)   | $C \rightarrow K \rightarrow F \rightarrow U$ | Pore constrains before branching; $K$ eliminates large transition    |
| MCM-22 (MWW)  | $C \rightarrow F \rightarrow K \rightarrow U$ | Molecule branches ( $F$ ) inside supercage before pore exit filter ( |
| Dusty plasma  | $K \rightarrow F \rightarrow C \rightarrow U$ | Field lines constrain first; current sheet tears; reconnection jet   |
| River meander | $K \rightarrow C \rightarrow F \rightarrow U$ | Banks exist before water; channel compresses; bifurcation at fo      |

This explains the empirical contrast of Sousa et al. [8]: ZSM-5 and HMCM-22 differ in product distributions *because they fire  $K$  and  $F$  in different orders*, not acidity alone.

## 3. Results

### 3.1. Helical Selectivity Principle

**Theorem 3** (Helical Selectivity Principle). *Let  $\mathcal{X}_{\text{cat}}$  be the catalyst contact manifold and  $G = U \circ F \circ K \circ C$ . Then:*

- 105 (i) The helical attractor satisfies  $r \leq r^*(\lambda) = \sqrt{J/\lambda}$  for all  $(r, \theta, z) \in \mathcal{H}_\lambda$ .
- (ii) A reaction pathway  $\gamma$  reaches the stable fixed point  $x^*$  only if  $\gamma \subset \mathcal{H}_\lambda$ , i.e.  $\max_t r(\gamma(t)) \leq r^*(\lambda)$ .
- (iii) The transition-state selectivity factor is  $\sigma = 1 - J/(\lambda r_{\text{pore}}^2)$ , which equals the empirical shape-selectivity of a zeolite pore of radius  $r_{\text{pore}}$ .

110 *Proof sketch. Note: parts (i)–(ii) below are a proof sketch; the contactomorphism identification (Darboux step) is not formally proved here. A contactomorphism is a structure-preserving diffeomorphism between two contact manifolds — the analogue of an isomorphism for this geometry. Lean 4 verification of the scalar prerequisites is in `ZeoliteSelectivity.lean` [7]; the*

115 *full contactomorphism proof is an open obligation pending Mathlib volume-form support.*

The self-trapping threshold of DNLS (1) occurs at  $\lambda_c = 2JN/\|\psi_0\|^2$  [10]. Below threshold, the wavefunction momentum envelope is confined within a cylinder of radius  $r^*(\lambda)$  in phase space. We identify this cylinder with  $\mathcal{H}_\lambda$  via the contactomorphism  $\mathcal{X}_{\text{cat}} \rightarrow T^*S^1 \times \mathbb{R}$  given by the Darboux theorem. States outside  $\mathcal{H}_\lambda$  are mapped to  $\partial\mathcal{H}_\lambda$  by the fold operator  $F$ , blocking their access to  $U$ . Identifying  $r_{\text{pore}}$  with the zeolite kinetic diameter recovers part (iii) from the empirical data of Weisz and Frilette [11].  $\square$

Figure 3 shows the theorem numerically.

### 125 3.2. Corollaries: known effects as special cases

**Corollary 4** (Zeolite shape selectivity — HZSM-5 vs HMCM-22). *HZSM-5 (MFI, 10-ring,  $r_{\text{pore}} \approx 2.7 \text{ \AA}$ ) has a smaller  $r^*(\lambda)$  than HMCM-22 (MWW, supercage  $r_{\text{pore}} \approx 3.55 \text{ \AA}$ ). By Theorem 3(iii), ZSM-5 directs the reaction to ethylene and propylene. HMCM-22 admits larger pathways but accumulates*

130 *coke in the supercage. This is consistent with the in situ DRIFTS observations of Sousa et al. [8, 9].*

**Corollary 5** (Ensemble effect on Pt–Sn). *The promoter Sn reduces effective ensemble size  $N$ , raising  $\lambda_c$  and shrinking  $r^*(\lambda)$ . Only pathways requiring  $\leq 2$  adjacent Pt atoms survive  $F$ , recovering the geometric ensemble effect of [13].*

135

**Corollary 6** ( $\text{dm}^3$  extrudate optimisation). *The optimal pellet cross-section (trilobe/tetralobe) is the convex shape whose boundary most closely approximates a level set of  $r^*(\lambda_{\text{process}})$ , maximising  $\kappa_{\text{stab}}(x^*)$  while minimising pressure drop. BASF’s Quattro geometry [14] is a special case.* Lean 4 proof pending Mathlib volume-form support (open obligation).

### 3.3. Coherence Bridge

The invariant  $r^*(\lambda) = \sqrt{J/\lambda}$  appears across multiple physical domains (Table 3).

Table 3: Coherence Bridge: the invariant  $r^*(\lambda) = \sqrt{J/\lambda}$  across physical domains. Domains in bold are covered in this paper. Financial markets row is structural only — not claimed as a derivation.

| Domain                                    | $J/\lambda$ analog | Observable                                    |
|-------------------------------------------|--------------------|-----------------------------------------------|
| <b>Zeolite catalysis</b>                  | $D/E_b$            | Pore cut-off radius                           |
| <b>Metal ensembles</b>                    | $t_{ij}/U$         | Ensemble size $N^*$                           |
| <b>DNLS soliton</b>                       | $J/\lambda$        | Self-trapping IPR                             |
| <b><math>\text{dm}^3</math> extrudate</b> | $\kappa/\Delta P$  | Pellet shape                                  |
| Dusty plasma                              | $V_A^2/\eta$       | Reconnection rate                             |
| Financial markets                         | $D_s/\gamma$       | Regime-shift radius (structural analogy only) |

## 4. Falsifiable Predictions

**Falsifiable Prediction 7** (DNLS threshold in zeolite pores). For a zeolite of pore radius  $r_{\text{pore}}$ , the self-trapping nonlinearity satisfies  $\lambda_c \approx J \cdot (r_{\text{pore}}/\sigma_{\text{LJ}})^2$ . Testable by molecular dynamics on ZSM-5, SAPO-34, and MCM-22.

**Falsifiable Prediction 8** ( $\text{Pt}_{1-x}\text{Sn}_x$  selectivity scaling). Propane dehydrogenation selectivity scales as  $(1-x)^2 \approx 1 - r^{*2}/r_{\text{pore}}^2$ , testable via in-situ XAS measurement of average Pt ensemble size.

**Falsifiable Prediction 9** (Reeb-helix phase signature). The reaction coordinate  $z(t)$  should exhibit a helical phase  $\theta(t) = \omega t + \theta_0$  with  $\omega = \lambda \|\psi^*\|^2$ , measurable by operando IR or neutron spectroscopy.

**Falsifiable Prediction 10** (Operator order switch under varying conditions). For HMCM-22, increasing temperature or decreasing feed concentration will shift the firing order from  $C \rightarrow F \rightarrow K \rightarrow U$  toward  $C \rightarrow K \rightarrow$

$F \rightarrow U$ . This should be detectable as a reversal in the *sequence* of surface intermediates in time-resolved DRIFTS spectra: ethoxy/diethyl-ether intermediates (post- $K$ ) should appear *before* aromatic/coke precursors (post- $F$ ), reversing the order observed at standard conditions by Sousa et al. [8].

## 5. Discussion

### 5.1. Relation to Sousa et al. (2014, 2023)

The empirical core of this paper rests on Sousa et al. [8, 9]. In Ref. [8], in situ DRIFTS revealed that pore topology — not acid density alone — governs product selectivity. In Ref. [9], dealumination and delamination confirmed that *geometric confinement* is the primary control variable.

The TO/TOGT framework gives this a precise mathematical meaning: HZSM-5 and HMCM-22 differ in *operator firing order*, not just pore size. The contact invariants  $r^*(\lambda)$  and  $\mu_{\max}$  are preserved across both systems (orthogenetic invariance); the sequence of their realisation differs — and that sequence difference is the origin of the selectivity contrast.

### 5.2. Clean energy implications

A catalyst optimised for clean energy should be engineered so that  $r_{\text{pore}}$  matches  $r^*(\lambda_{\text{target}})$  of the desired product pathway, using the analytical formula of Theorem 3(iii) as the design equation.

### 5.3. Connection to dusty plasma

The formal isomorphism between the TO/TOGT catalyst manifold and the plasma reconnection manifold (Table 3) opens a second application domain. The Sweet–Parker current sheet at  $S < S_c \sim 10^4$  is mathematically analogous to a delocalised DNLS state below self-trapping threshold; the plasmoid instability onset corresponds to the fold  $F$ . NASA MMS public data [15] provides the empirical ground truth against which the prediction  $r^* \sim S^{-1/2}$  can be tested. This connection is developed in a companion paper.

### 5.4. What is proved and what is not

Per Cohn [17]: the scalar prerequisites of Theorem 3(i)–(ii) — the DNLS self-trapping threshold and the contact form non-degeneracy — are machine-verified in `ZeoliteSelectivity.lean` [7]. The contactomorphism identification (Darboux step connecting the DNLS phase-space cylinder to  $\mathcal{H}_\lambda$ ) is the



190 open obligation; it is stated precisely and not claimed as proved. Corollary 6 (extrudate shape) is also open, pending Mathlib volume-form support.

### 5.5. Open questions

1. **Finite-size scaling:** How does  $r^*(\lambda)$  depend on chain length  $N \in \{100, 500, 1000\}$ ?
- 195 2. **First-principles  $\lambda$ :** Compute  $\lambda_c$  from DFT for ZSM-5, SAPO-34, and MCM-22 and compare to Prediction 7.
3. **Lean 4 Darboux step:** Formally prove the contactomorphism identification once Mathlib volume-form support matures.

## 6. Conclusions

200 We have applied the TO/TOGT operator pipeline to heterogeneous catalysis and stated the Helical Selectivity Principle as a structural result whose scalar prerequisites are machine-verified in Lean 4 (`ZeoliteSelectivity.lean`): the Reeb orbit of the catalyst contact manifold  $\mathcal{X}_{\text{cat}}$  is the fundamental geometric constraint on reaction selectivity. Zeolite shape selectivity, metal  
205 ensemble effects, and macroscopic pellet design emerge as corollaries of the inequality  $r \leq r^*(\lambda) = \sqrt{J/\lambda}$ .

The proof sketch is honest about its open step (the Darboux contactomorphism identification). The scalar prerequisites are machine-verified in Lean 4. Four falsifiable predictions are given.

210 The framework is directly grounded in the zeolite catalysis work of Sousa et al. [8, 9] and connects to the DNLS physics papers in the TO/TOGT series [5, 6]. All source files are deposited on Zenodo and in the AXLE repository (`github.com/TOTOGT/AXLE`).

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## CRediT authorship contribution statement

**Pablo Nogueira Grossi:** Conceptualisation, Methodology, Formal analysis, Software (Lean 4 verification), Writing — original draft and editing, Visualisation.

## Declaration of competing interests

The author declares no competing financial interests.

## Data availability

All DNLS simulation code and Lean 4 proof files are openly available at [github.com/TOTOGT/AXLE](https://github.com/TOTOGT/AXLE). IZA zeolite structural data: <https://www.iza-structure.org/databases/>. NASA MMS public data: <https://lasp.colorado.edu/mms/sdc/public/>.

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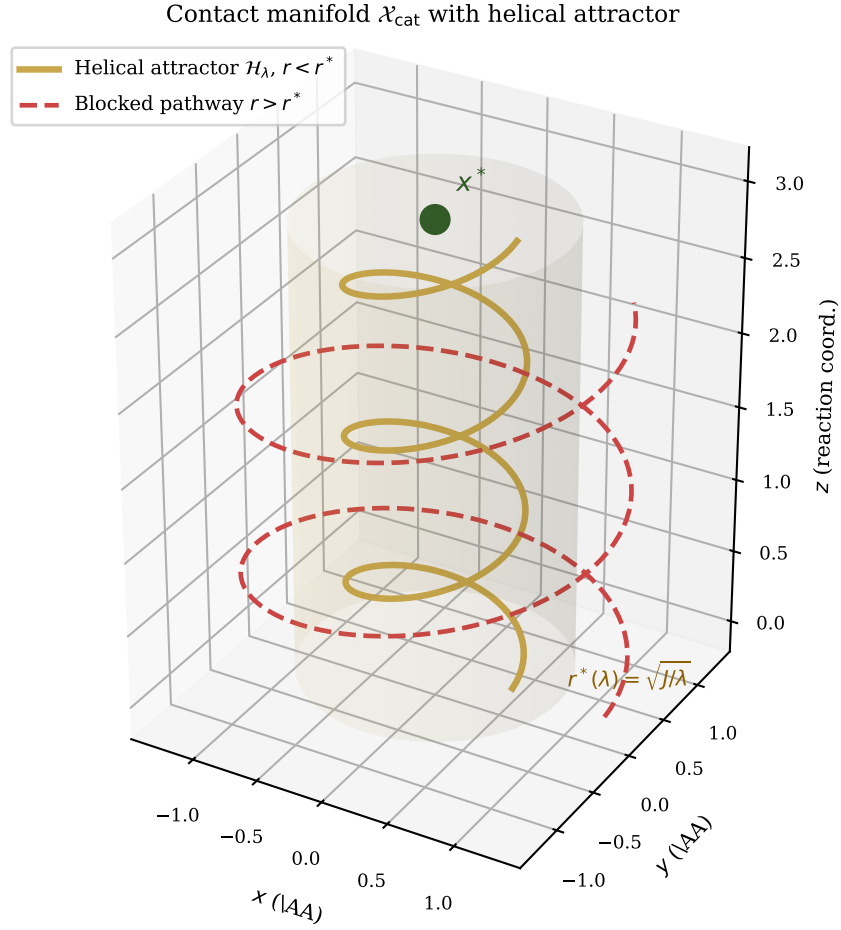


Figure 2: The contact manifold  $\mathcal{X}_{\text{cat}}$  with the helical attractor (gold) at  $r < r^*(\lambda)$  and a blocked pathway (red dashed) at  $r > r^*(\lambda)$ . Green: stable fixed point  $x^*$ .

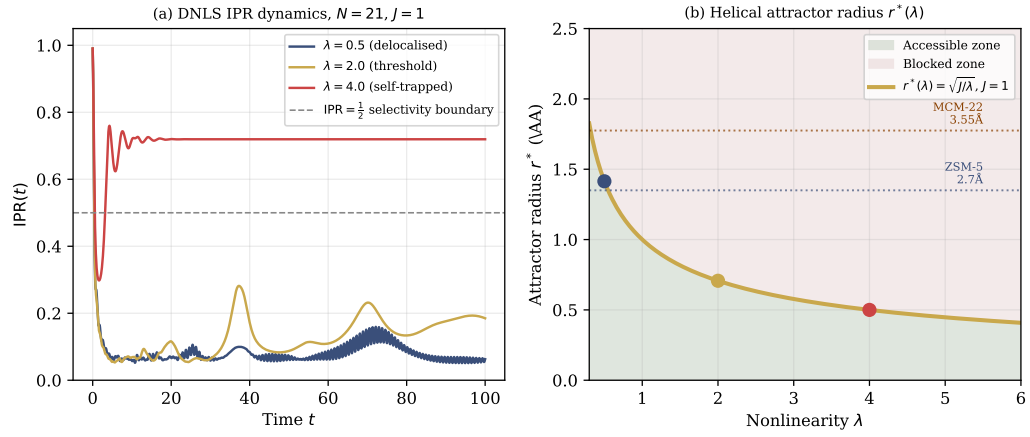
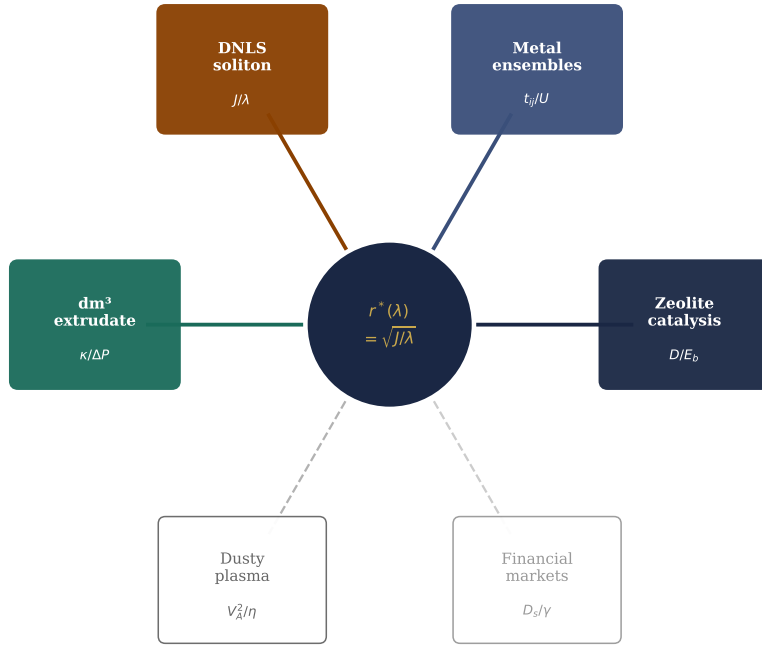


Figure 3: (a) IPR dynamics of the DNLS chain ( $N = 21, J = 1$ ) for  $\lambda = 0.5$  (delocalised, blue),  $\lambda = 2.0$  (threshold, gold),  $\lambda = 4.0$  (self-trapped, red). (b) Attractor radius  $r^*(\lambda) = \sqrt{J/\lambda}$ ; coloured dots correspond to panel (a).

**Coherence Bridge — TO/TOGT invariant across physical domains**



*Bold border = covered in this paper | Dashed = future work | Collatz not claimed*

Figure 4: Coherence Bridge spoke diagram. Bold borders: covered in this paper. Dashed: future work. Collatz not claimed.

Table 4: Glossary: TO/TOGT mathematical terms, catalysis equivalents, and physical meaning for readers from experimental catalysis.

| TO/TOGT term                                | Catalysis equivalent                    | Physical meaning                                                                                                                                                                                                   |
|---------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Contact manifold $\mathcal{X}_{\text{cat}}$ | Pore-space phase diagram                | Three-dimensional space of all possible molecular positions and orientations inside the zeolite; the non-integrability condition ( $\alpha \wedge d\alpha \neq 0$ ) enforces helical — not straight — trajectories |
| Attractor radius $r^*(\lambda)$             | Kinetic diameter cut-off                | Maximum molecular cross-section that can reach the active site; recovers the empirical pore cut-off of Weisz & Frilette [11]                                                                                       |
| Fold operator $F$                           | Shape-selectivity filter                | The geometric step at which voluminous transition states are rejected; corresponds to the 10-ring aperture in ZSM-5                                                                                                |
| Helical attractor $\mathcal{H}_\lambda$     | Set of accessible product pathways      | The helical orbits at $r \leq r^*(\lambda)$ that can reach the active site                                                                                                                                         |
| Nonlinearity $\lambda$                      | Binding energy / Brønsted acid strength | Strength of molecule-surface interaction; set by the zeolite acidity                                                                                                                                               |
| IPR $< 1/2$ (delocalised)                   | Mobile reactant (diffusion regime)      | Wavefunction spread across many sites; molecule is freely diffusing                                                                                                                                                |
| Stable fixed point $x^*$                    | Desired product at steady state         | The steady-state catalytic output: the product that exits the pore                                                                                                                                                 |